

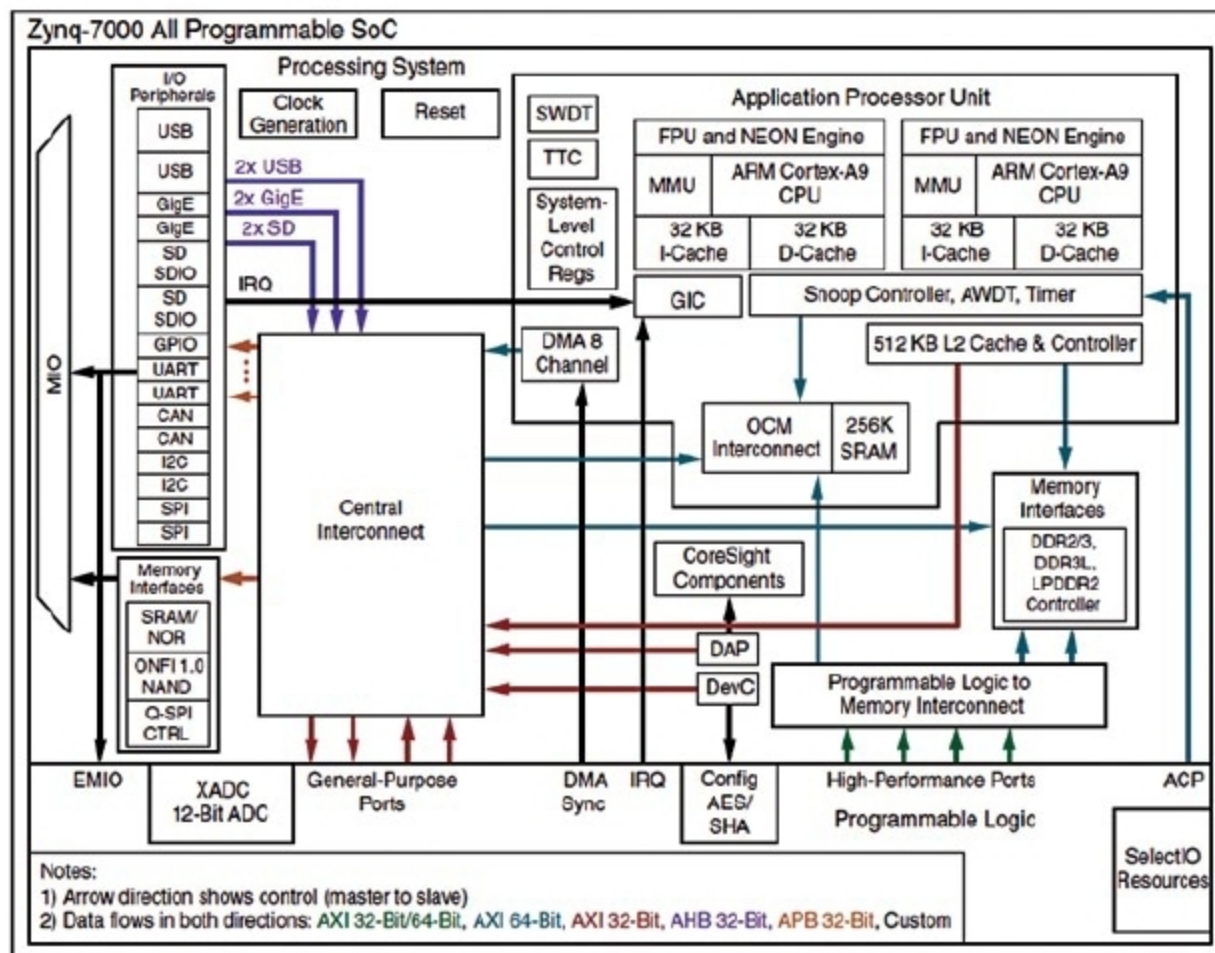


Fig. 2: Zynq SoC block

can be transferred to the FPGA or be distributed among PC and FPGA processors.

A simple pseudo code example below shows how the virtual node for a discrete cosine transform (DCT) hardware IP block is handled. The IP block is given a unique node number at the time of design; 1 in this case. By default, MCAPI send_msg function calls the driver of DMA controller (C-macro HPD_send), but for this virtual node number, it forwards execution to a DCT driver. This eventually calls HPD_SEND function, perhaps multiple times, to configure the DCT and send raw data to be transformed.

The same procedure is easily repeated for other accelerators.

Pseudo code.

```
send_msg(...) {
    ...
    switch(node_id){
    case '1' :
        ...
        dct_drv(...);
        ...
    default :
        ...
        HPD_SEND(...);
        ...
    }
}
```

Functions

Connection-less messages and packet channel API functions have blocking and non-blocking variants. For example, mcapi_msg_rcv(...) function blocks execution of the application node until all message data has been received. Non-blocking variants return immediately and, hence, their names are denoted with i.

Required buffers are statically reserved at compile time just like end-points and channels to increase overall reliability and ease of debugging. Applications must take care of data alignment, endian-ness and the internal structure of messages.

API of the framework has an elaborate structure and is very well-organised. It gives programmers multiple ways to access the services offered by the framework. Each access to the framework is made by using tasks.

Task. A task corresponds to a single processing operation. It is associated with one or many inputs that are available on different channels. If all inputs associated with a task are activated (available), then that task is activated and is queued up for execution.

An input option marked as final is made available. If a task has an input

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INSULATION TESTERS



Model	Range	Test Voltage DC
MC901BA	0-20 M Ohms	100 V
MC903BA	0-100 M Ohms	500 V
MC904BA	0-500 M Ohms	500 V
MC941BA	0-1000 M Ohms	500 V
MC906BA	0-200 M Ohms	1000 V
MC907BA	0-500 M Ohms	1000 V
MC981BA	0-2000 M Ohms	1000 V



Model	Range	Test Voltage DC	Resolution
DIT 99A	0-20 M Ohms	100 V	0.01 M Ohms
DIT 99B	0-200 M Ohms	250 V	0.1 M Ohms
DIT 99C	0-200 M Ohms	500 V	0.1 M Ohms
DIT 99D	0-200 M Ohms	1000 V	0.1 M Ohms
DIT 99E	0-2000 M Ohms	1000 V	1 M Ohms



Specification	Test Voltage	Range
Insulation Resistance	100V / 250V / 500V	0.1MΩ to 200GΩ
AC Voltage Measurement	0-100V AC (50-60Hz)	-
Phase Sequence Test	100V-450V (Phase-Phase) 40-60Hz	-

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